Reference Guide: LaTeX Tabular Environment Parameters

Reference Guide

Introduction

The `tabular` environment is used to create tables in LaTeX. Its basic syntax is:

```
\begin{tabular}[pos]{cols}
 table content
\end{tabular}
```

This guide provides a comprehensive reference for all parameters and options available in the `tabular` environment.

1 Position Parameters (`pos`)

The `pos` parameter determines the vertical alignment of the table relative to the surrounding text baseline.

Value	**Description**
`t`	The top line of the table is aligned with the text baseline.
`b`	The bottom line of the table is aligned with the text baseline.
`c` or *omitted*	The table is centred vertically relative to the text baseline (default).

Table 1: Position parameters for the tabular environment

2 Column Formatting Parameters (`cols`)

The `cols` parameter defines the alignment, width, and borders for each column.

Value	Description
`l`	Left-justified column.
`c`	Centred column.
`r`	Right-justified column.

Table 2: Basic column alignment options

Value	Description
`p{width}`	Paragraph column with text vertically aligned at the top.
`m{width}`	Paragraph column with text vertically aligned in the middle (requires `array` package).
`b{width}`	Paragraph column with text vertically aligned at the bottom (requires `array` package).

Table 3: Paragraph column options

Basic Column Alignment

Paragraph Columns

Border Formatting

Column Repetition

3 Table Content Commands

Commands used within the `tabular` environment to structure and format content.

4 Complete Reference Summary

Parameter	Type	Description
`pos`	`t`	Aligns the top of the table with the text baseline.
	`b`	Aligns the bottom of the table with the text baseline.
	`c`	Centres the table vertically relative to the text baseline (default).
	omitted	Equivalent to `c`; table is centred vertically.
`cols`	`l`	Left-justified column.
	`c`	Centred column.
	`r`	Right-justified column.
	`p{width}`	Paragraph column, top-aligned text.
	`m{width}`	Paragraph column, middle-aligned text (requires `array`).

Table continues on next page

Parameter	Type	Description
	`b{width}`	Paragraph column, bottom-aligned text (requires `array`).
`cols` (borders)	`|`	Single vertical line.
	`||`	Double vertical line.
	`*{num}{form}`	Repeats `form` `num` times.
	Example:	`*{3}{|l|}` = `|l|l|l|`.
Commands	`&`	Column separator.
	`\\`	New row (optional spacing with `[...]`).
	`\hline`	Full-width horizontal line.
	`\newline`	New line within a paragraph column cell.
	`\cline{i-j}`	Partial horizontal line from column `i` to `j`.

5 Usage Examples

Example 1: Basic Table

```
\begin{tabular}{|l|c|r|}
 \hline
 Left & Centre & Right \\
 \hline
 A & B & C \\
 D & E & F \\
 \hline
\end{tabular}
```

Example 2: With Fixed Width Columns

```
\usepackage{array}  % in preamble

\begin{tabular}{|p{3cm}|m{4cm}|b{3cm}|}
 \hline
 Top-aligned paragraph & Middle-aligned paragraph & Bottom-aligned paragraph \\
 \hline
 This is a longer text that will wrap within the cell. & This text will wrap and be
 \hline
\end{tabular}
```

Example 3: With Column Repetition

```
\begin{tabular}{*{3}{|c}|}
 \hline
 1 & 2 & 3 \\
```

Value	Description
`|`	Single vertical line between columns or at the edges.
`||`	Double vertical line between columns or at the edges.

Table 4: Border formatting options

Value	Description
`*{num}{form}`	Repeats the format `form` exactly `num` times. For example, `*{3}{|l|}` is equivalent to `|l|l|l|`.

Table 5: Column repetition syntax

```
  4 & 5 & 6 \\
  \hline
\end{tabular}
```

Example 4: Complex Table with Borders

```
\begin{tabular}{|c||c|c||c|}
  \hline
  \multicolumn{1}{|c||}{Header 1} &
  \multicolumn{2}{c||}{Header 2} &
  \multicolumn{1}{c|}{Header 3} \\
  \hline
  A & B & C & D \\
  \cline{2-3}
  E & F & G & H \\
  \hline
\end{tabular}
```

Important Notes

- The `array` package is required for `m{width}` and `b{width}` column types.
- The `booktabs` package provides enhanced horizontal line commands (`\toprule`, `\midrule`, `\bottomrule`) for professional-looking tables.
- Column widths should be specified using LaTeX length units: `cm`, `mm`, `in`, `pt`, `em`, `ex`, `linewidth`, `textwidth`, etc.
- The `*{num}{form}` syntax can be nested in complex column definitions.

Package Dependencies

Command	Description
&	Column separator; moves to the next column in the current row.
\\	Starts a new row. Additional vertical space can be specified in square brackets, e.g., \\[6pt].
\hline	Draws a horizontal line across the full width of the table.
\newline	Starts a new line within a cell (only valid in paragraph columns).
\cline{i-j}	Draws a partial horizontal line from column i to column j.

Table 6: Table content and formatting commands

Package	Functionality Provided
array	Adds m{width} and b{width} column types; extends column specification capabilities.
booktabs	Provides enhanced horizontal line commands for professional table formatting.
multirow	Enables cells to span multiple rows.
colortbl	Adds colour support for rows, columns, and individual cells.
longtable	Extends tabular to support multi-page tables.

Table 8: Packages that extend tabular functionality